WORKED SOLUTION

STEP I 1990, Question 3

Pure • Curves

STEP I 1990, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q3

The key idea

Everything follows once we write down the downward unit normal explicitly and so get the coordinates of $\mathbf{s}$ in terms of x. Part (i) is then a matter of forcing the y-coordinate of $\mathbf{s}$ to vanish. In part (ii) the suggested substitution makes the coordinates of $\mathbf{s}$ very simple, and the gradient condition turns into a first-order differential equation that turns out to be an exact derivative.

Solution

Setting up. The tangent direction to $y = \mathrm{f}(x)$ is $(1, \mathrm{f}'(x))$, so the normals are $\pm(\mathrm{f}'(x), -1)$. The downward one has negative y-component, so

$$\mathbf{n} = \frac{1}{N}\begin{pmatrix} \mathrm{f}'(x) \\ -1 \end{pmatrix}, \qquad N = \sqrt{1 + \mathrm{f}'(x)^2},$$

and therefore

$$\mathbf{s} = \mathbf{r} + \mathrm{p}(x)\mathbf{n} = \begin{pmatrix} x + \dfrac{\mathrm{p}(x)\,\mathrm{f}'(x)}{N} \\ \mathrm{f}(x) - \dfrac{\mathrm{p}(x)}{N} \end{pmatrix}.$$

Note also that $|\mathbf{r} - \mathbf{s}| = |\mathrm{p}(x)|$, since $\mathbf{n}$ is a unit vector.

Part (i)

Here $\mathrm{f}(x) = x^k$, $\mathrm{f}'(x) = kx^{k-1}$ and $N = \sqrt{1 + k^2 x^{2k-2}}$. For $\mathbf{s}$ to lie on the x-axis we need its y-coordinate to be zero:

$$x^k - \frac{\mathrm{p}(x)}{N} = 0 \quad \Longrightarrow \quad \mathrm{p}(x) = x^k\sqrt{1 + k^2 x^{2k-2}}.$$

With this choice the x-coordinate of $\mathbf{s}$ is

$$x + x^k \cdot kx^{k-1} = x + kx^{2k-1},$$

which is continuous, increasing, tends to 0 as $x \to 0^+$ and tends to infinity as $x \to \infty$. So $\mathbf{s}$ really does sweep out the whole positive x-axis, not just part of it.

Now $|\mathbf{r}|^2 = x^2 + x^{2k}$, and $|\mathbf{r} - \mathbf{s}|^2 = \mathrm{p}(x)^2 = x^{2k} + k^2 x^{4k-2}$ (here $\mathrm{p} > 0$). So $|\mathbf{r}| = |\mathbf{r} - \mathbf{s}|$ for every $x > 0$ exactly when

$$x^2 = k^2 x^{4k-2} \quad \text{for all } x > 0.$$

Taking $x = 1$ forces $k^2 = 1$, so $k = 1$; and conversely $k = 1$ makes both sides equal to x^2. Hence $k = 1$.

Geometrically: on the line $y = x$, the downward normal from (x, x) meets the x-axis at $(2x, 0)$, and the triangle with vertices O, (x, x), $(2x, 0)$ is isosceles with both equal sides of length $\sqrt{2}\,x$.

Part (ii)

Now $\mathrm{f}(x) = x^2$, $\mathrm{f}'(x) = 2x$ and $N = \sqrt{1 + 4x^2}$. Writing $\mathrm{p} = \mathrm{q}N$ cancels the awkward square roots:

$$\mathbf{s} = \begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} x + 2x\,\mathrm{q}(x) \\ x^2 - \mathrm{q}(x) \end{pmatrix}.$$

The new curve is given parametrically by x, so its gradient at $\mathbf{s}$ is

$$\frac{\mathrm{d}Y}{\mathrm{d}X} = \frac{\mathrm{d}Y/\mathrm{d}x}{\mathrm{d}X/\mathrm{d}x} = \frac{2x - \mathrm{q}'}{1 + 2\mathrm{q} + 2x\mathrm{q}'}.$$

Setting this equal to $2x$, the gradient of $y = x^2$ at $\mathbf{r}$, and multiplying up:

$$\begin{aligned} & 2x - \mathrm{q}' = 2x + 4x\mathrm{q} + 4x^2\mathrm{q}' \\ \iff \quad & (1 + 4x^2)\,\mathrm{q}' + 4x\,\mathrm{q} = 0. \end{aligned}$$

> **Ask yourself** Can we solve this without dividing by q (which might be zero somewhere)?

Yes: divide instead by $N = \sqrt{1 + 4x^2}$, which is never zero. Since $N' = 4x/N$,

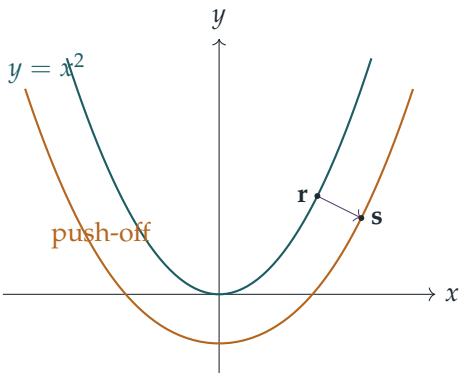
$$\frac{(1 + 4x^2)\,\mathrm{q}' + 4x\,\mathrm{q}}{N} = N\mathrm{q}' + \frac{4x}{N}\,\mathrm{q} = \frac{\mathrm{d}}{\mathrm{d}x}(N\mathrm{q}) = \mathrm{p}'(x).$$

So the condition is simply $\mathrm{p}'(x) = 0$ for all x, and hence

$$\mathrm{p}(x) = c, \qquad \text{a constant}$$

(equivalently $\mathrm{q}(x) = c/\sqrt{1 + 4x^2}$). Conversely, every constant c works, since each step above is reversible wherever the gradient of the new curve is defined.

The new curve is a *parallel curve*: every point of the parabola is pushed a fixed distance c along its normal. The picture shows $c = \frac{1}{2}$.



One subtlety: $\mathrm{d}X/\mathrm{d}x = 1 + 2\mathrm{q} + 2x\mathrm{q}'$, which with $\mathrm{q} = c/N$ simplifies to $1 + 2c/N^3$. For $c > -\frac{1}{2}$ this is always positive, so the push-off is a smooth curve. If $c \leq -\frac{1}{2}$ (pushing *upwards* by at least $\frac{1}{2}$, the radius of curvature at the vertex), it vanishes where $N^3 = -2c$ and the push-off develops cusps there.

Checks

- In (i) with $k = 2$ and $x = 1$: $\mathrm{p}(1) = \sqrt{5}$, and $\mathbf{s} = (1,1) + \sqrt{5} \cdot \frac{1}{\sqrt{5}}(2,-1) = (3,0)$, on the x-axis as required; but $|\mathbf{r}| = \sqrt{2} \neq \sqrt{5}$, so $k = 2$ fails.
- In (ii), at $x = 0$ the push-off with constant c passes through $(0,-c)$, and by symmetry its gradient there is 0, matching the parabola's vertex.
- Numerically, for $c = 1$ and $x = 0.8$, a finite-difference gradient of the push-off agrees with $2x = 1.6$; with the non-constant choice $\mathrm{p} = x^2$ it comes out as about 0.41 instead.

Common slips

- Using the upward normal $(-\mathrm{f}', 1)/N$: the signs in $\mathbf{s}$ all flip.
- Writing the gradient of the new curve as $\mathrm{d}Y/\mathrm{d}x$ rather than $(\mathrm{d}Y/\mathrm{d}x)/(\mathrm{d}X/\mathrm{d}x)$; the new curve's x-coordinate is X, not x.
- In (i), concluding $k = 1$ from matching exponents ($4k - 2 = 2$) without also checking the coefficient $k^2 = 1$.

Beyond STEP

Push-offs at constant distance are called *parallel* or *offset* curves, and they matter in practice: a milling machine's cutter centre follows the parallel curve of the shape being cut. The cusps that appear when the offset exceeds the radius of curvature lie on the *evolute*, the locus of centres of curvature, a central object in the differential geometry of

curves. A good introduction is Pressley, *Elementary Differential Geometry*.

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